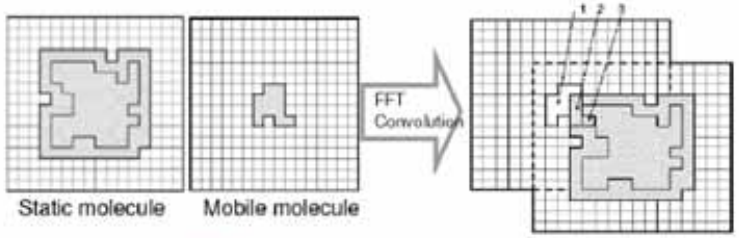
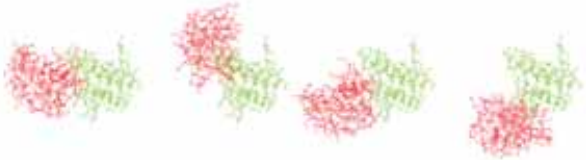
Protein-protein rigid-body docking methods

Exhaustive search
(FFT, surface-based)





Scoring function

- Energy terms: ES, H-bonds, ...
- Conservation
- Statistical potentials

Geometry-based docking

Stochastic sampling
(Monte-Carlo, minimization)

- Geometry-based docking
  - **FFT-based grid search**
  - **Geometric Hashing**

FTDOCK

```
ftdock -static 1A2P_A.pdb -mobile 1A19_A.pdb
        -noelec -calculate_grid 1.2 -angle_step 12
        -internal -15 -surface 1.3 -keep 3
        -out 1BRS.ftdock > 1BRS.ftdock.log
```

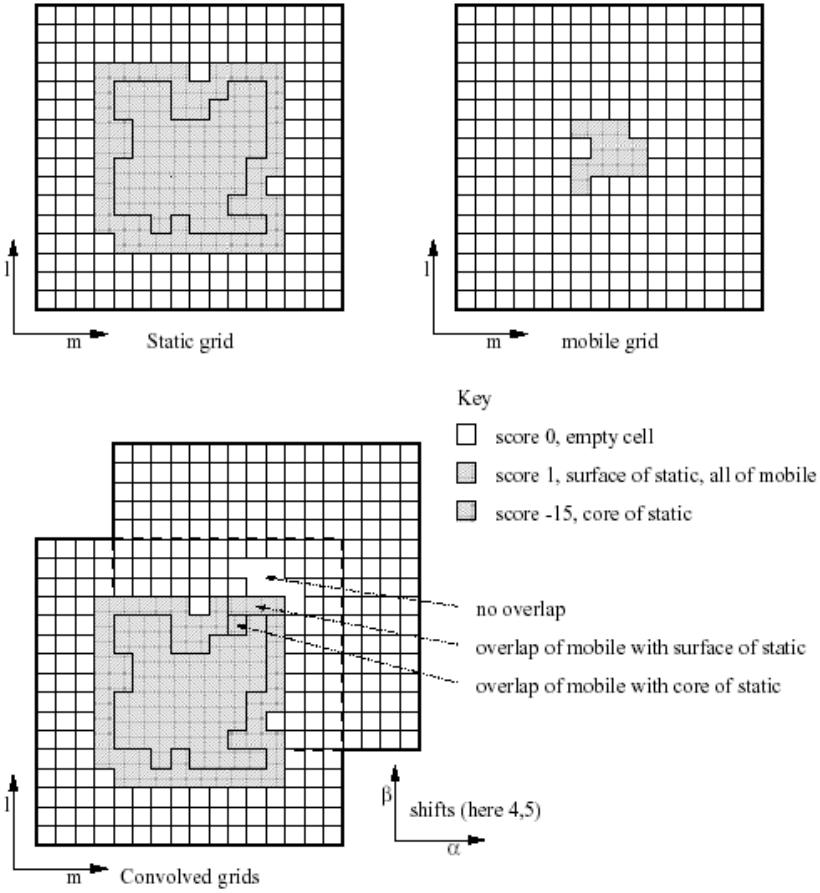
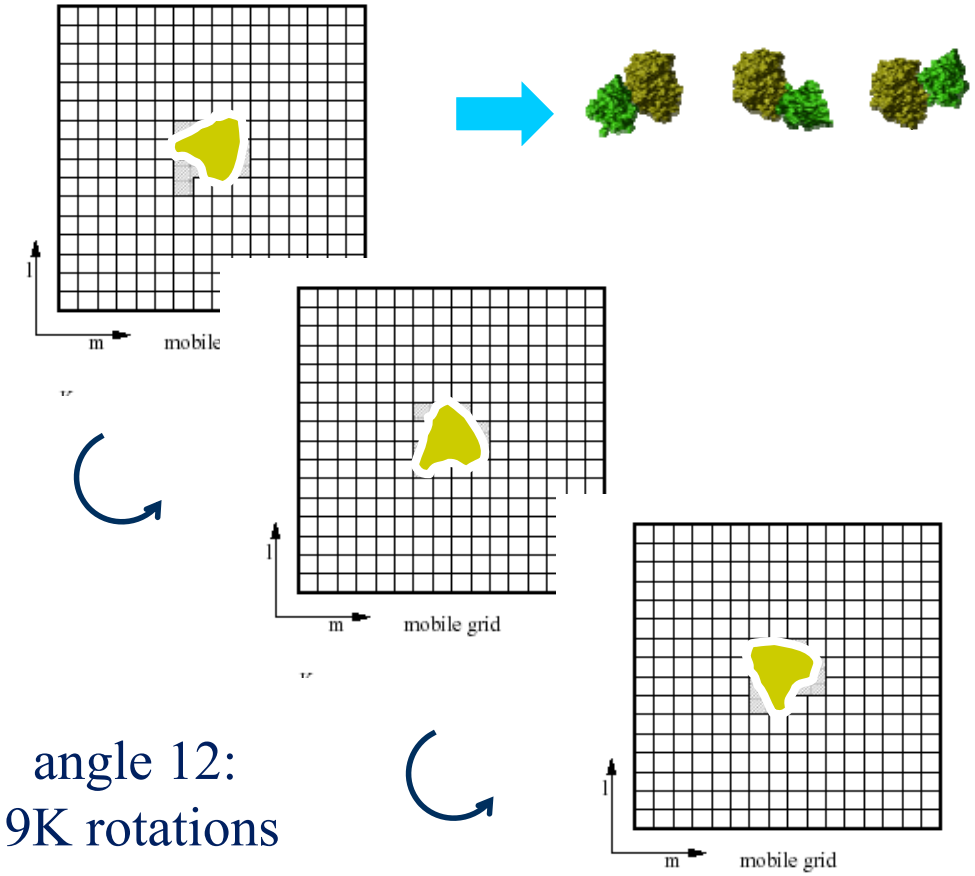
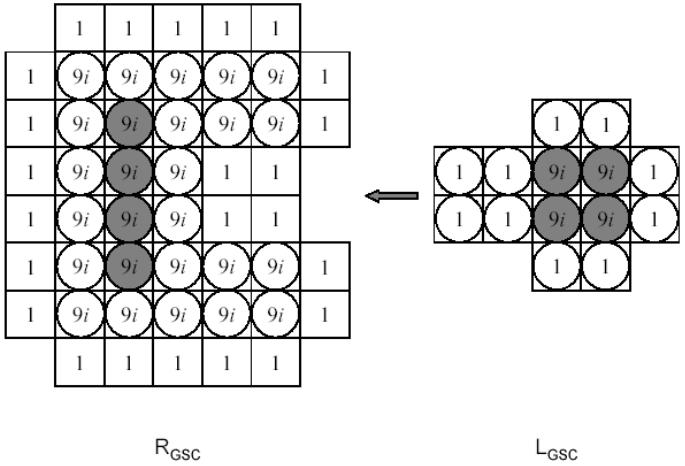


Figure 2: Grid discretisation of molecules and calculation of surface complementarity

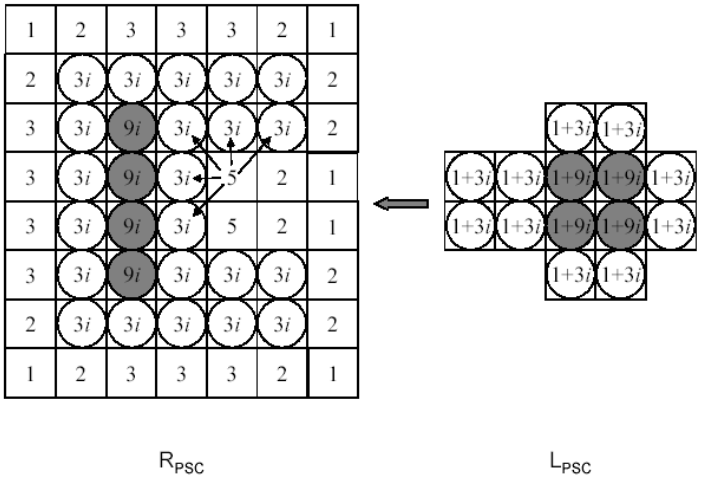


angle 12:
9K rotations

ZDOCK



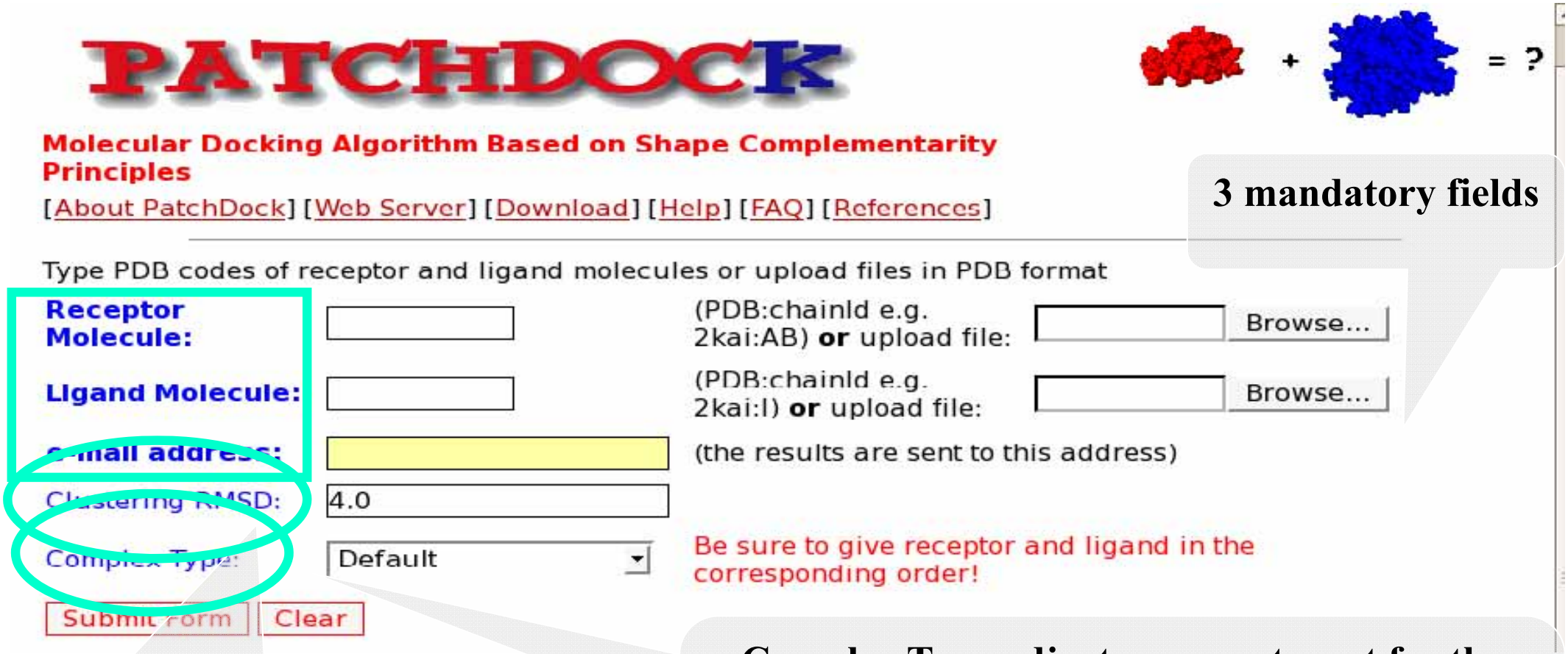
mark_sur 1A2P_A.pdb 1BRS_r.pdb
mark_sur 1A19_A.pdb 1BRS_l.pdb



zdock -R 1BRS_r.pdb -L 1BRS_l.pdb -o 1BRS_zdock.out

- Geometry-based docking
  - **FFT-based grid search**
  - **Geometric Hashing**

WWW server: http://bioinfo3d.cs.tau.ac.il/PatchDock



3 mandatory fields

Clustering RMSD determines the number of output solutions

Complex Type adjusts parameter set for the specific complex type, such as enzyme-inhibitor, antibody-antigen, protein-small ligand

Schneidman-Duhovny D, Inbar Y, Nussinov R, Wolfson HJ. NAR 2005

1) Run docking on example 1

ZDOCK , FTDock

2) Analysis of results

Example 1: Create PDBs, visualize results,
compare with real complex structure...

3) Run docking on example 2

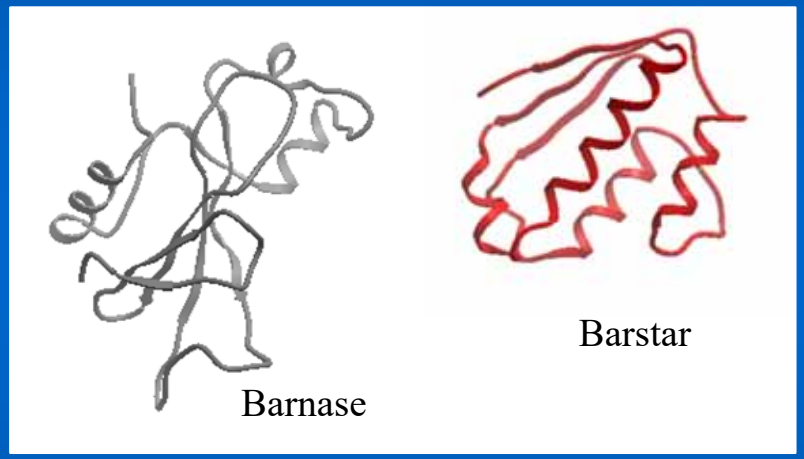
PatchDock

4) Analysis of results

Example 2: Create PDBs, visualize results,
compare with real complex structure...

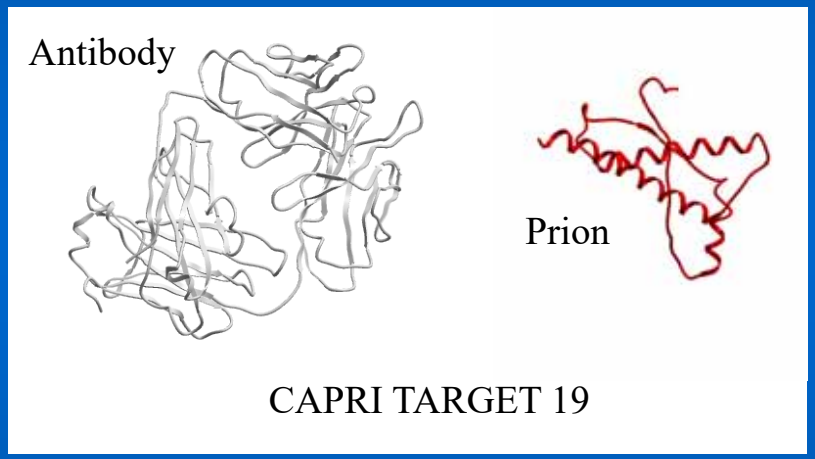
Example 1

ZDOCK (server and local)
FTDock (local)



Example 2

PatchDock (local)



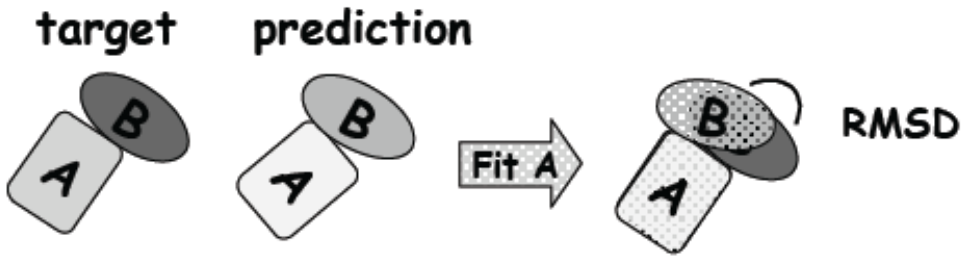
Assessment of docking performance

- F_{nat}: fraction of native contacts (within 5Å)

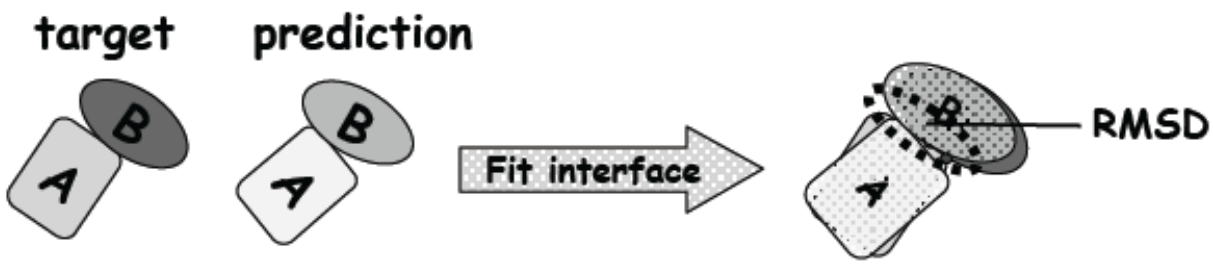
$$F_{nat} = \frac{\text{Correctly predicted contacts}}{\text{Total number of contacts in the target}}$$

- l-RMSD: RMSD on second protein after superposition on first

near-native:
L-RMSD < 10Å



- i-RMSD: RMSD on interface residues (within 10 Å)



Assessment of docking performance

Rank	Scoring	L-RMSD
1	2569	34.9
2	2456	14.8
3	2019	5.6
...	...	...
...	...	...
10	1567	23.5